Install light fitting with reflectors for direct and indirect lightings

Objectives: At the end of this exercise you shall be able to
- **design the light reflectors to a given room according to working situation**
- **install and check the effect of light reflection.**

Requirements

Tools/Instruments

- Cutting plier 200mm - 1 No.
- Screw driver 150mm - 1 No.
- Drilling machine electric 6 mm capacity with drill bit - 5 mm - 1 No.

Materials

- Incandescent lamp 100W 240V with lamp shades of similar design - 2 No.
- Reflector lamp 100W 240V - 2 No.
- Silvered bowl lamp 100W 240 - 2 No.
- Wiring materials - as reqd.

PROCEDURE

1 Identify the loaction and working situation, for light reflectors.

2 Carry out the marking for fixing the two ceiling roses at close proximity.

3 Carry out the wiring as per the circuit.

4 Check the wiring with a series test board.

5 Hang one lamp shade to face down, and one lamp shade up side down as shown in Fig 1 with suitable strings.

6 Fix similar type (incandescent) bulbs in both the shades.

7 Keep some display articles just beneath the lamp shades on a table.

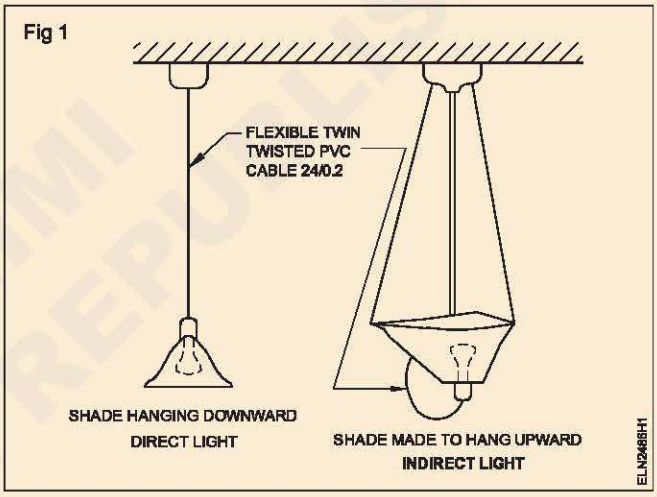
8 Give the supply and check the lighting emitted by direct and indirect lights one by one.

9 Check the illumination of the lighting emitted by direct and indirect lights one by one.

10 Change the bulbs of similar variety (say reflector type) of same wattage in the shades and repeat the steps 8 and 9.

11 Write the conclusion based on the illumination level and suitability of direct and indirect lighting of displaying mercantize.

Fig 1



Conclusion :

_____

_____

_____

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Group different wattage lamps in series for specified voltage

Objectives: At the end of this exercise you shall be able to
- **read and interpret the data stamped on a given lamp**
- **measure the voltage drop across the lamp when unequal wattage lamps are connected in series to the supply**
- **state the reasons for the behaviour/condition of glow of unequal wattage lamps in series.**

Requirements

Tools/Instruments

- Multimeter - 1 No.
- Voltmeter MC 0-15V - 3 Nos.
- Ammeter MC 0-500 mA - 1 No.

Equipment/Machines

- DC variable source 0-24 volts, 5 amps with output current & voltage indicator - 1 No.

Materials

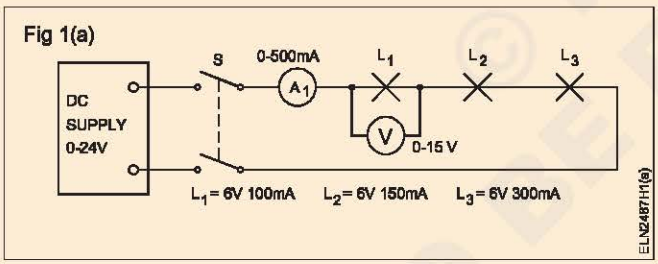
- Bulbs screw cap - 6V 100 mA - 10 Nos.
- Bulbs screw cap - 6V 150 mA - 6 Nos.
- Bulbs screw cap - 6V 300 mA - 4 Nos.
- Bulb-holders - 20 Nos.
- Connecting leads - as reqd.
- Knife switch DPST 16A - 1 No.

PROCEDURE

TASK 1 : **Connect 3 lamps of 6 volts in series across 18 volts supply (unequal wattage) and test it**

1 Connect the three lamps with ammeter A in series to the variable voltage DC supply source Fig 1a.

Fig 1(a)



Keep the output of DC source at minimum, say 0 volts.

2 Connect a MC voltmeter (0-15 V) across L_1 (i.e low current rating/low wattage bulb). Close the switch S.

3 Gradually increase the supply voltage from 0 volts, observing ammeter, voltmeter and lamp L_1.

4 Increase the voltage upto 18 volts. Record your observations.

_____

_____

_____

5 Does the lamp L_1 fuse? If yes, give your reasons, stating the observation made just before fusing.

_____

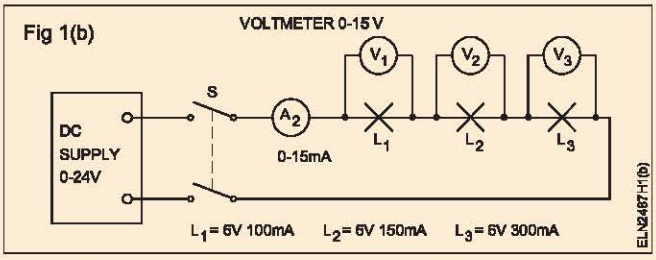
_____

_____

6 Open the switch S and reset the supply voltage to OV. Replace the bulb L_1.

7 Form the circuit Fig 1(b) with 3 voltmeters 0-15 volts connected across each lamp.

Fig 1(b)



8 Close the switch S and increase the supply voltage until the current reaches 100 mA., (i.e. rated current of low wattage bulb in the series circuit).

9 Read the voltages V_1, V_2 & V_3 and record in Table 1.

Table 1

Supply Voltage	V_1	V_2	V_3

10 Give your reasons for the unequal distribution of supply voltage.

11 Connect each lamp L_1, L_2 & L_3 independently in the circuit Fig 2 and record the value of current and voltage when the supply voltage is 6 V in Table 2.

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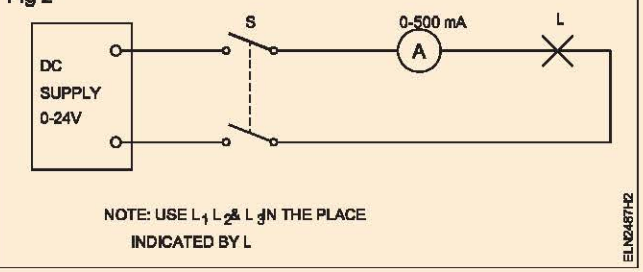
Table 2

Lamp in circuit	Supply voltage	V	I	V/I
L_1 6 V 100 mA	6 V			
L_2 6 V 150 mA	6 V			
L_3 6 V 300 mA	6 V			

Fig 2



Conclusion

The voltage across each of the lamps connected in series varied because of

_____

_____

_____

The stamped value of voltage and current on the lamp means that the specified _____ when applied will cause a _____ to flow.

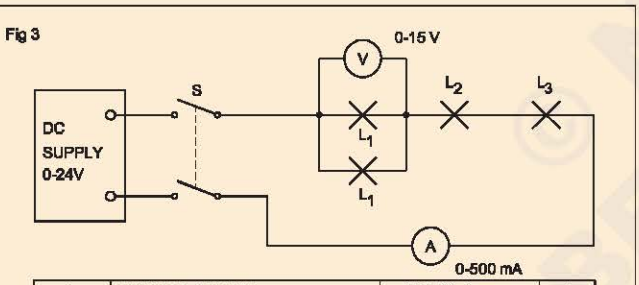
Resistance of lamp varies because of different _____ of lamp.

— — — — — — — —

TASK 2 : **Connect two low wattage L_1 lamps in parallel as in Task 1 and test it**

1 Form the circuit as per the diagram, Fig 3.

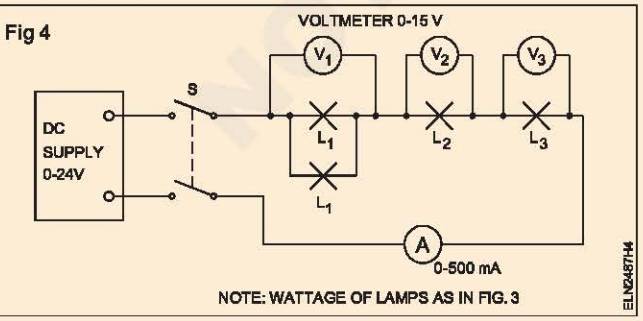
Fig 3



L_1	MINIATURE LAMP	6V,100mA	2
L_2	MINIATURE LAMP	6V,150mA	1
L_3	MINIATURE LAMP	6V,300mA	1

2 Observe the effect of increasing the supply voltage from 0 to a value that causes 6 volts across low wattage lamps L_1.

Fig 4



3 Does the lamp L_2 fuse? If yes, give your reasons stating observations made just at the time of L_2 fusing.

_____

_____

_____

4 Open the switch S, connect the 3 voltmeters as shown in Fig 4.

5 Replace the lamp L_2 and reset the DC source at OV. Close the switch S. Increase the supply voltage until a current of 150 mA flows in the circuit.

6 Read and record the voltages V_1, V_2 & V_3 in Table 3.

Table 3

Supply Voltage	V_1	V_2	V_3

Conclusion

The voltage is V_2 is greater than V_1. Also $V_2 > V_3$, because

_____

_____

_____

— — — — — — — —

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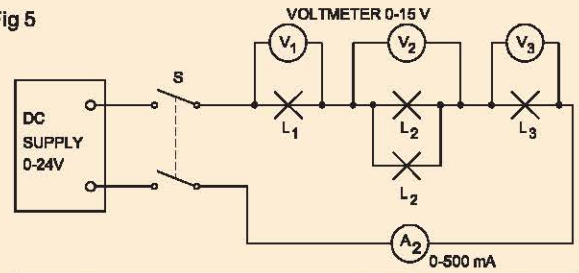
TASK 3 : **Connect two (L_2 lamps) in parallel as in task 1 and test it**

1 Form the circuit as shown in Fig 5.

Fig 5



L_1	MINIATURE LAMP	6V,100mA	1
L_2	MINIATURE LAMP	6V,150mA	2
L_3	MINIATURE LAMP	6V,300mA	1

ELN2497-5

2 Gradually increase the supply voltage up to 18V after closing the switch S. Observe the voltage V_1, current and glow of lamp L_1.

3 Does the lamp L_1 fuse again? What are the conditions at the time of fusing?

_____ V_1
_____ DC supply voltage
_____ current

4 Is there any difference in the values as compared with the one in Task 1? Give your response.

5 Open the switch S. Replace the fused lamp L_1. Reset the supply voltage to 0V. Close the switch S and increase the current through the circuit to 100 mA. Record the voltages V_1, V_2 & V_3 in Table 4.

Table 4

Supply Voltage	V_1	V_2	V_3

Conclusion

_____

_____

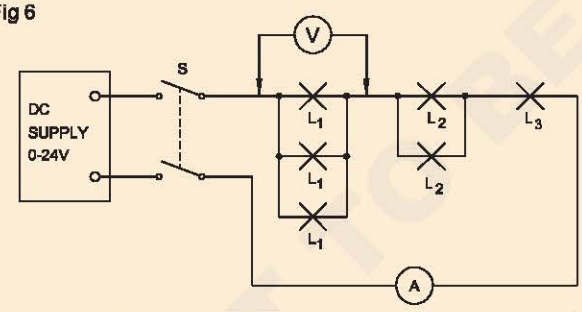
_____

— — — — — — — —

TASK 4 : **Connect three L_1 lamps connected in parallel and the whole in series with one lamp L_3 two L_2 lamps in parallel as in task 1**

1 Form the circuit as shown in Fig 6.

Fig 6



L_1	MINIATURE LAMP	6V,100mA	3
L_2	MINIATURE LAMP	6V,150mA	2
L_3	MINIATURE LAMP	6V,300mA	1

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2 Close the switch S. Increase the supply voltage gradually to 18 V. Observe the lamps, ammeter and measure the voltage across the lamp group L_1, lamp group L_2 and L_3.

3 Now all the lamps glow with their normal brightness. No lamp fused. Why?

_____

_____

_____

Conclusion

In a serial set of lamps, while replacing a fused lamp the lamps voltage and also _____ or _____ should also _____ wattage lamp, replacement should be

— — — — — — — —

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Practice installation of various lamps eg. fluorescent tube, HP mercury vapour, LP mercury vapour, HP Sodium vapour, LP Sodium vapour, Metal halide etc.

Objectives: At the end of this exercise you shall be able to
- **connect a flourescent tube with accessories, install and test it**
- **connect a H.P. M.V lamp with acccessories, install and test it**
- **connect a H.P.S.V lamp with accessories install and test it**
- **connect a L.P.S.V lamp with accessories install and test it**
- **connect a metal halide lamp with accessories install and test it.**

Scan the QR Code to view the video for this exercise

Requirements

Tools/Instruments

- Insulated combination plier - 150 mm - 1 No.
- Insulated screwdriver - 200 mm x 4mm - 1 No.
- Insulated connector screwdriver - 100 mm - 1 No.
- Long round nose plier - 150 mm - 1 No.
- D.B. Electrician's knife 100 mm - 1 No.
- Test lamp 100 W, 250 V - 1 No.

Materials

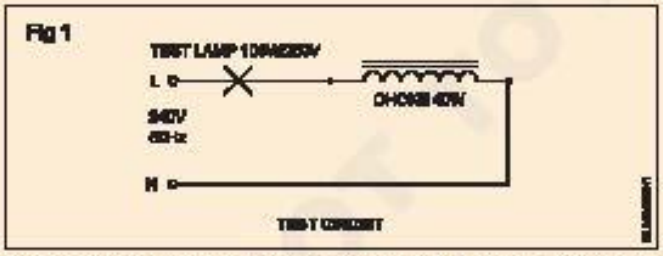
- Tube light fitting 1200 mm - single patti - 1 No.
- Choke 40w, 250 V - 1 No.
- Tube light starter - 40W, 250V - 1 No.
- Tube light holder plain - 2 Nos.
- Starter holder - 2 Nos.
- MV lamp holder suitable for 240W, 250 V lamp (Goliath screw type) - 2 Nos. single patti - 1 No.
- MV lamp choke - 240 Watts, 250 V - 1 No.
- Capacitor 4 MFD / 380 U - 1 No.
- L.P.M.V lamp 40 W, 250 V - 1 No.
- MV lamp 240W, 250V - 1 No.

PROCEDURE

TASK 1: **Assembling of a fluorescent lamp (LPMV lamp) with its accessories**

1 Check the choke for its short and open with a test lamp as shown in Fig 1.

Fig 1



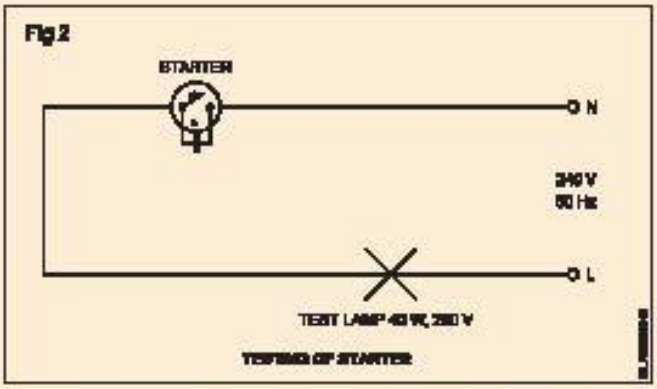
2 Check the starter with a series test lamp as shown in Fig 2. Observe the flickering of the lamp which indicates good condition of the starter.

3 Assemble the following fluorescent tube accessories in the fitting base. Refer to the sketch. (Fig 3)

1) Holders for tube 2) Starter-holder 3) Choke.

4 Connect the accessories as shown in Fig 4 (for a single tube light). Also install the tested starter.

Fig 2

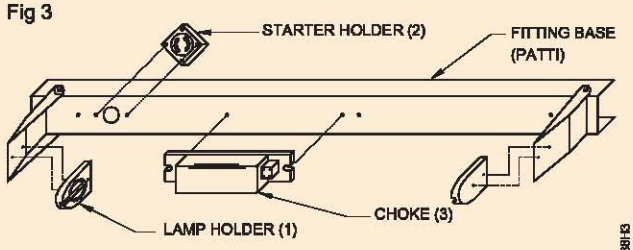


5 Test the filament on both sides of the fluorescent tube for its continuity as shown in Fig 5. Discard the fluorescent tube with open or fused filament in either side.

6 Fix the bulb in the holder.

Firstly, you have to make sure that the slot in the inner parts of the holder is turned to the proper position.

Fig 3



ASSEMBLY OF ACCESSORIES

Fig 4

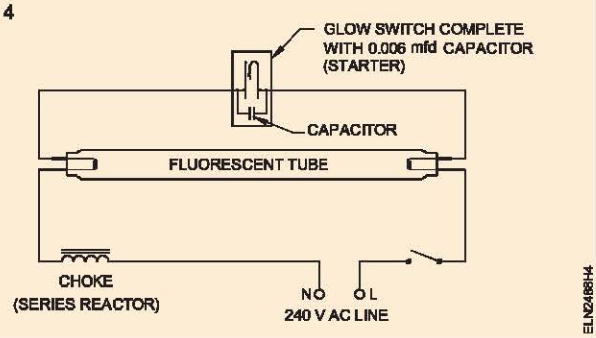
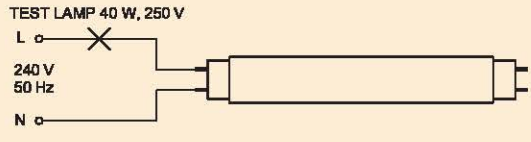


Fig 5



TESTING THE TUBE

7 Test the tube light assembly for its working.

TASK 2 : **Installation of tube light fitting**

1 Follow the recommended method and procedure depending on the type of wiring.

The fixing of the tube to the wall, ceiling or tubular post should be strong enough to support the weight of the fitting.

The installed fitting must be below the level of the ceiling fan to avoid the flickering effect of the shadow.

2 Connect the tube light fitting to the ceiling rose.

Check the supply at the ceiling rose. Switch off the supply before making any connection.

3 Fix the fluorescent tube in the fitting.

Use a stable ladder and a helper to hold the ladder while you are working on the ladder.

4 Switch `ON' the supply and observe the glow of the tube. If the tube is not glowing, check for proper housing of starter and tube.

TASK 3 : **Install and test the H.P.M.V (High Pressure Mercury Vapour) lamp with accessories**

1 Read the specification of the mercury vapour lamp and the choke from the markings. (Fig 6)

2 Connect the H.P.M.V. lamp in series with the 60W 240V bulb and test in 240V AC supply. Check whether the series test lamp glows.

3 Test the choke for its working condition.

4 Assemble the accessories (choke, holder and capacitor) in the fitting, following the manufacturer's instructions.

5 Connect the accessories as per circuit diagram, Fig 7 (Pictorial diagram Fig 8) using the recommended type of termination.

Choose the tapping of the choke suitable to the rated supply system voltage.

6 Fix the bulb in the holder and test the working of the lamp with the supply voltage.

Ensure the fitting is properly earthed at the earthing terminal provided, before testing.

7 A modern M.V. lamp with a built-in resistor needs no external accessories to be connected as discussed above. It can be connected as we do an incandescent lamp.

Installation of the M V lamp fitting

8 Assemble, connect and test the M.V. lamp fitting on a table, for its working. Then remove the cover and bulb.

Mount at the location

9 Observe the recommended method and procedure specified by the manufacturer in the installation leaflet.

Do not alter the specifications recommended by the manufacturer because it should be strong enough to support the weight of the fitting.

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Fig 6

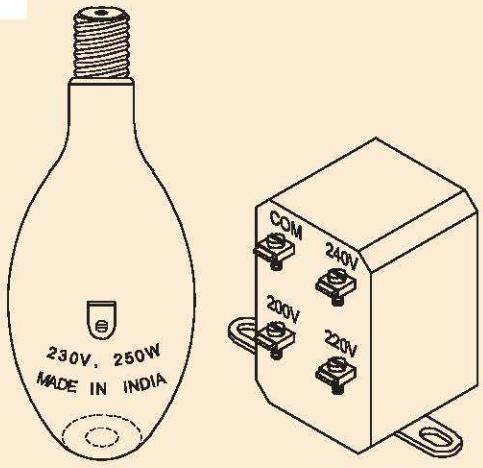
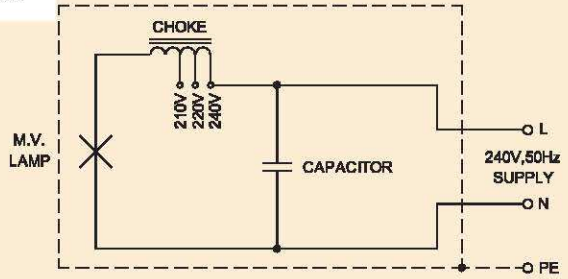


Fig 7



CIRCUIT DIAGRAM OF HPMV LAMP

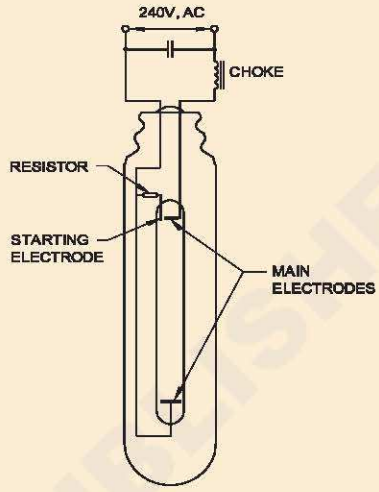
10 Connect the M.V. lamp fitting to the supply. The method depends on the system of wiring, location of fitting etc.

Ensure that the supply line is dead (not live), before making the connections.

11 Fix the bulb in the holder securely and refit the cover.

12 Switch on the supply and wait until the high pressure mercury vapour lamp glows with its full brightness. then switch off the supply.

Fig 8



PICTORIAL DIAGRAM OF A HPMV LAMP

TASK 4 : **Install and test H.P.S.V. (High Pressure Sodium Vapour) and LPS lamp with accessories**

1 Read the specification from the markings on the leak transformer, choke and bulb.

2 Check the transformer and choke with a test lamp for shorts and open.

3 Assemble the accessories (choke, leak transformer and lamp-holder) in the fitting.

Follow strictly the manufacturer's instructions.

4 Give connections as per diagram shown in Fig 9.

Use the recommended type of termination only.

5 Choose the appropriate voltage tapping suitable to the supply voltage. (Fig 9)

6 Fix the bulb in the holder.

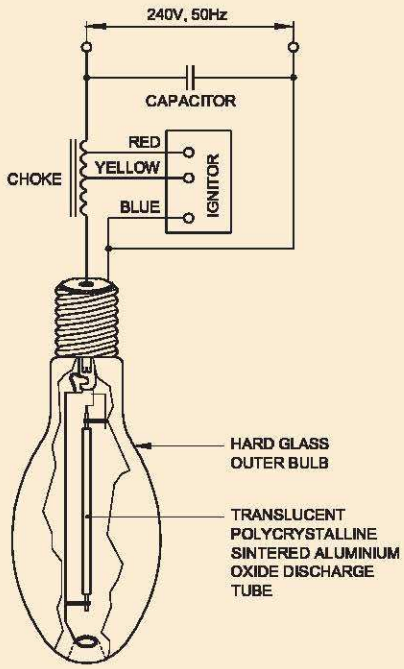
Ensure the fitting is properly earthed.

7 Test the working of the assembled fitting by connecting it to the mains.

8 Note the time taken for the bulb to give full illumination.

9 Repeat the above steps for a high pressure sodium vapour lamp. Connect as per the diagram shown in Fig 10.

Fig 9



HIGH PRESSURE SODIUM VAPOUR LAMP

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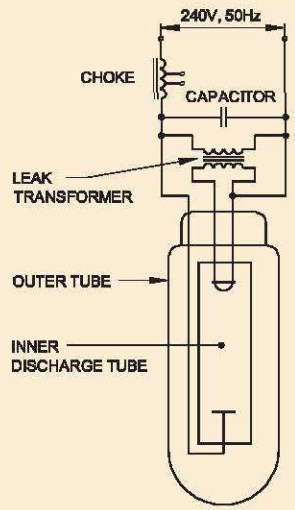
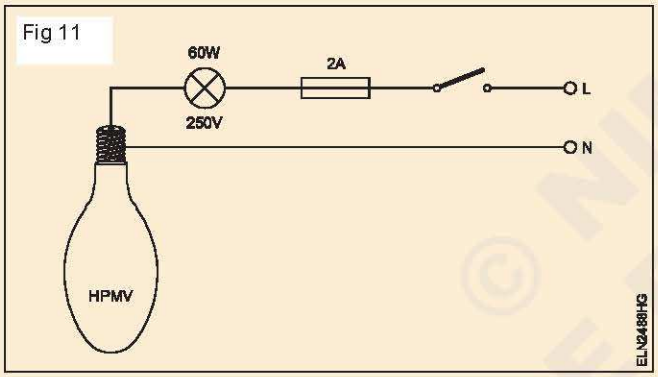
Fig 10



TASK 5 : **Testing of High prssure metal Halide**

1 Read the specifications of the given Halide lamp as Fig 11 collect the required accessories.

Fig 11



2 Connect the HPMH lamp in sries with a 60W. 250V incandescent amp as shown in fig. 11 and test with 240V AC supply. Check whether the series test lamp glows. If the test lamp flows it means that HPMV lamp in good codition.

3 Connect as the circuit diagram and test with 240V supply.

4 Measure the current and test with 240V supply.

Measure the current and voltage. Calculate the power and verify with the rated values.

Voltage :_____Volt

Current :_____Amp

Power :_____Watt

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Prepare a decorative lamp circuit to produce rotating light effect/ running light effect

Objectives: At the end of this exercise you shall be able to
- **select lamps/sequential control for light decoration**
- **design lighting layout for running light**
- **design layout for rotating light**
- **connect the motor for 3-point running light (sequential control motor)**
- **connect lamp circuits in the electronic sequential controller.**

Requirements

Tools/Instruments
- Multimeter - 1 No.

Equipment/Machines
- Single phase motor FHP with reduction gear - 1 No.
- 240V operation output load 5 to 10 A with speed and intensity control - 2 Nos.

Materials
- Cams - 3 Nos.
- Brushes - 3 Nos.
- Connection leads flexible - as reqd.
- Cam drive arrangement with shaft - 1 No.
- Lamps 240V, 15W, BC - 54 Nos.
- Batten Lamp holder 6A, 250 V - 54 Nos.
- DPST knife switch 16A 250V - 2 Nos.
- Electronic sequential controller - 1 No.

PROCEDURE

TASK 1 : **Prepare a rotating**

1 Connect the lamps, switches and the flasher motor. (Fig 1).

2 Keep the D.P.S.T switches S_1 & S_2 Open.

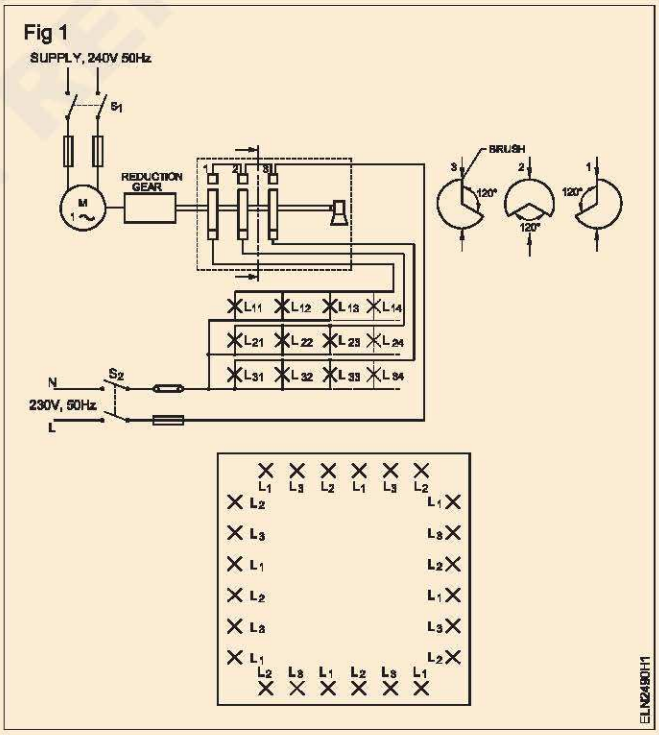
3 Close the D.P.S.T switch S_1 and start the flasher motor (sequential light controller.

4 Close the D.P.S.T Switch S_2 and observe the make and break contacts 1,2,3 and "ON" "OFF" operations of the 3 lamp banks.

Do not touch live wires

6 Open the D.P.S.T. switch S_1 and S_2

Fig 1



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TASK 2 : **Prepare a running light effect**

1 Prepare the lighting design as shown in Fig 2.

2 Close the D.P.S.T. switch S_1 and observe the lighting.

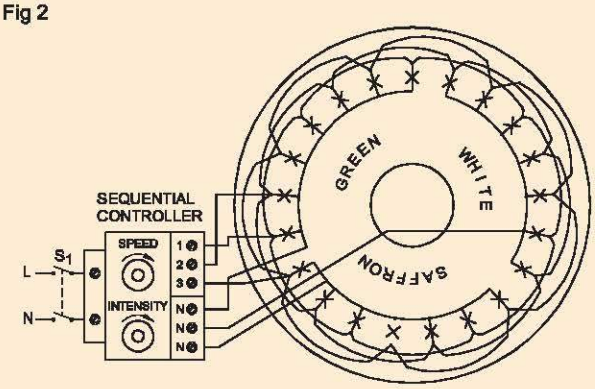
3 Increase the speed of operation by operating the speed control.

4 Adjust the intensity of light-adjusting the knob on the electric sequential controller.

5 Reduce the speed and intensity of the lighting system.

6 Open the D.P.S.T. switch S_1.

Fig 2



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Install light fitting for show case lighting

Objectives: At the end of this exercise you shall be able to
- **install and wire up the show case window lighting for tie rack**
- **wireup a show case window lighting to display clothes.**

Requirements

Tools and Instruments

- Insulated cutting pliers 150 mm - 1 No.
- Screw driver set of five - 1 Set
- Line tester 500V - 1 No.
- Electric hand drilling machine 6 mm capacity - 1 No.

Materials

- Complete set of circline tube light 30 cm 32 watts 250V 50 Hz with suitable shade and stand - 1 No.
- Complete set of 1200 mm fluorescent lamp fitting 40 watts 250V 50 Hz - 4 Nos.
- Wiring materials - as reqd.

PROCEDURE

TASK 1 : **Install and wire up the show case window lighting for tie rack**

1 Place a suitable sized plywood board in the base of the window with spacers.

2 Locate the circline tube fitting with its stand in proper position in the show case so that complete stand is visible from the window. Refer Fig.1.

3 Wire up in such a way that a 3 pin 5 amps socket is fitted in the inner side of the window.

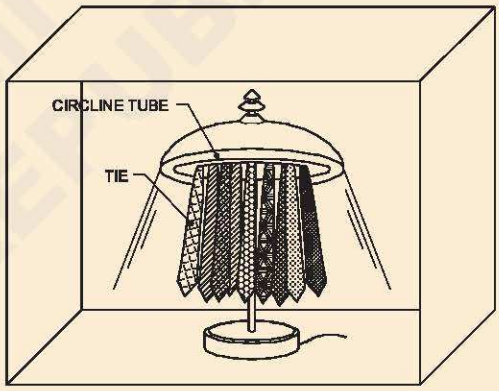
4 Mark the position of stand base and drill a hole in the marked centre to allow the circline tube cable to pass.

5 Draw the cable through the hole and connect a 3 pin plug at the cable ends.

6 Check the connections and connect the plug to the socket.

7 Give supply and check the lighting for the tie rack.

Fig 1



SHOWCASE WINDOW LIGHTING FOR TIE RACK

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TASK 2 : **Wire up the show case window lighting for a mannequin (dummy figure used for to display clothes)**

The show case needs four (400mm) tube light fittings to be wired in parallel and the tubes are hidden behind the frame. Refer Fig 2. Draw the connection diagram and wire up the fluorescent tubes in concealed wiring.

1 Prepare suitable frame for 4 tube light fittings which are to be hidden behind the frame (Fig 2)

2 Draw the connection diagram and wireup the 4 tube lights in parallel.

3 Place the dummy figure at the centre used to display the clothes

4 Get the supply and check for its functioning.

Fig 2

1,2,3 AND 4 TUBELIGHTS
TUBES AND WIRING ARE CONCEALED IN THE FRAME

INDIRECT LIGHTING OF SHOWCASE WINDOW

ELN2491H2

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